Timing Attack on MAC Verification

1. Aim

To demonstrate a timing attack on MAC verification and its prevention using a constant-time comparison function.

2. Description

This experiment demonstrates how an insecure comparison function leaks timing information, allowing attackers to guess MAC values. A secure comparison function is then used to eliminate this vulnerability.

3. Insecure Comparison Results

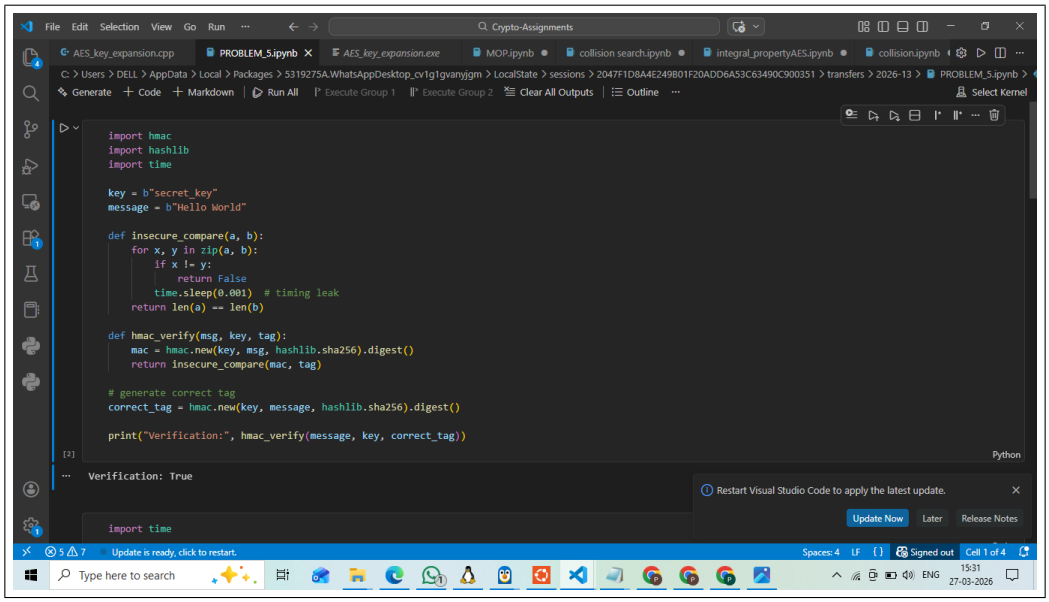

Figure 1: Initial insecure verification

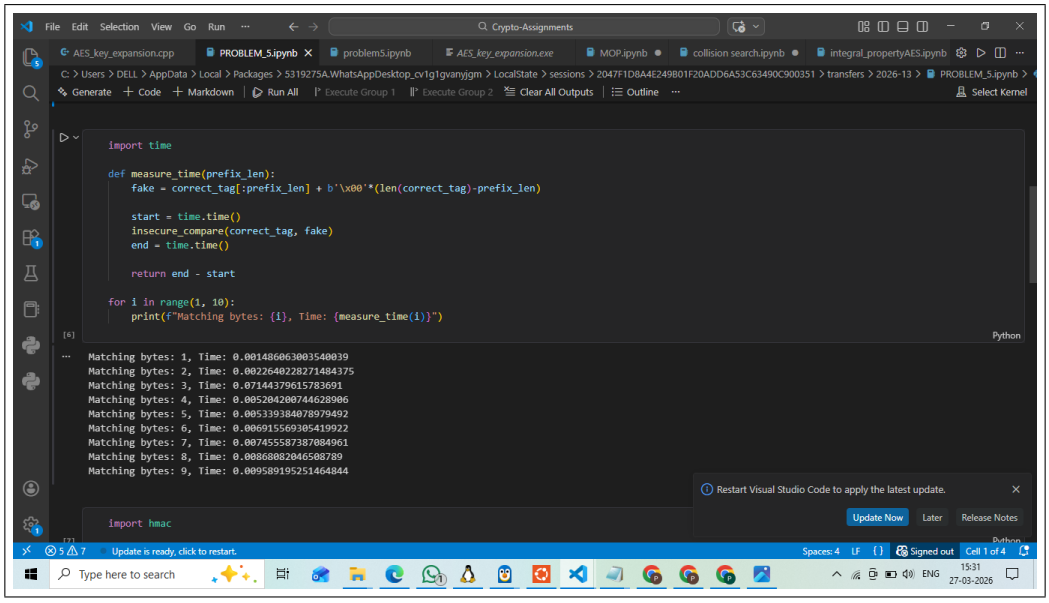

Figure 2: Timing increases with matching bytes

Observation:
The execution time increases as more bytes match, revealing a timing side-channel vulnerability.

4. Secure Comparison Results

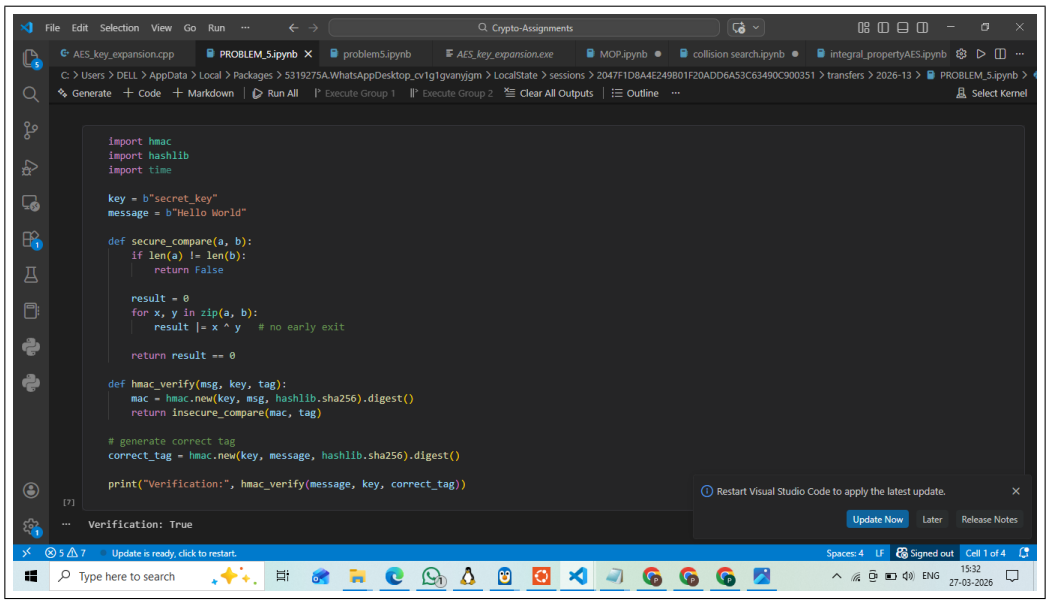

Figure 3: Verification using secure comparison

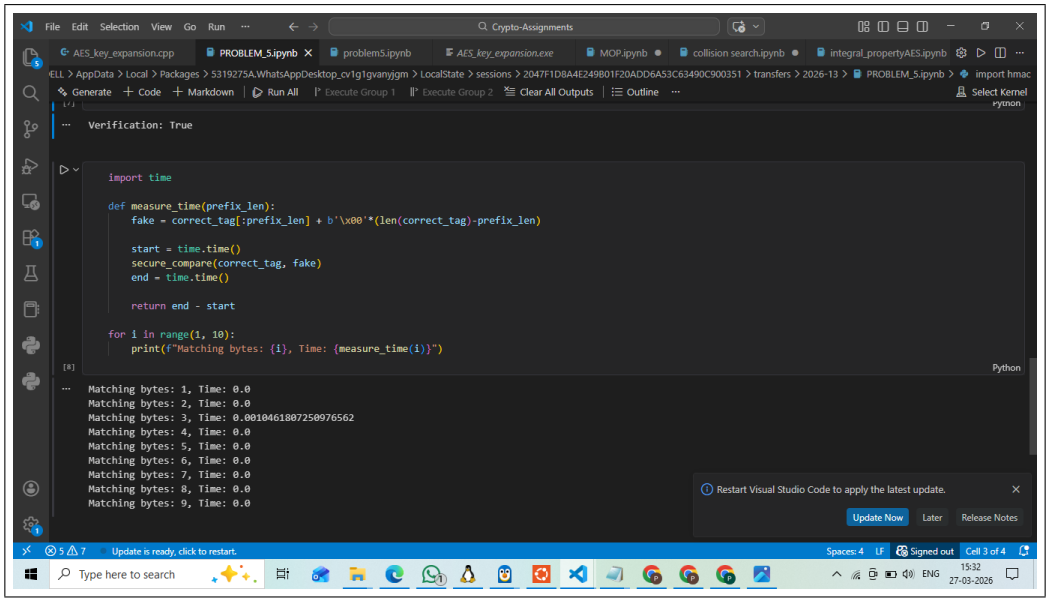

Figure 4: Constant execution time

Observation:
The execution time remains constant, preventing timing attacks and improving security.